

First Fit and Best Fit - C Programs

1. First Fit

```
#include <stdio.h>

int main() {
    int b[10], p[10], nb, np, i, j;

    printf("Enter number of blocks: ");
    scanf("%d", &nb);

    printf("Enter number of processes: ");
    scanf("%d", &np);

    printf("Enter block sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d", &b[i]);

    printf("Enter process sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d", &p[i]);

    for(i=0;i<np;i++) {
        for(j=0;j<nb;j++) {
            if(b[j] >= p[i]) {
                printf("Process %d allocated to block %d\n", i+1, j+1);
                b[j] -= p[i];
                break;
            }
        }
    }

    return 0;
}
```

Sample Input / Output

Sample Input:
Enter number of blocks: 3
Enter number of processes: 3
Enter block sizes:
100 500 200
Enter process sizes:
90 150 100

Sample Output:
Process 1 allocated to block 1
Process 2 allocated to block 2
Process 3 allocated to block 2

2. Best Fit

```
#include <stdio.h>

int main() {
    int b[10], p[10], nb, np, i, j, best;

    printf("Enter number of blocks: ");
    scanf("%d", &nb);

    printf("Enter number of processes: ");
    scanf("%d", &np);

    printf("Enter block sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d", &b[i]);

    printf("Enter process sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d", &p[i]);

    for(i=0;i<np;i++) {
        best = -1;

        for(j=0;j<nb;j++) {
```

```

        if(b[j] >= p[i]) {
            if(best == -1 || b[j] < b[best])
                best = j;
        }
    }

    if(best != -1) {
        printf("Process %d allocated to block %d\n", i+1, best+1);
        b[best] -= p[i];
    }
}

return 0;
}

```

Sample Input / Output

Sample Input:
 Enter number of blocks: 4
 Enter number of processes: 3
 Enter block sizes:
 100 500 200 300
 Enter process sizes:
 90 150 100

Sample Output:
 Process 1 allocated to block 1
 Process 2 allocated to block 3
 Process 3 allocated to block 4